Practice Material: https://docs.google.com/document/d/1iBB0qw32X-3nGlKcYcpWWIX0X0gQtUf4DEGzCzC9CLs/edit?usp=sharingFinal Exam Contents

Total Duration : 1 hour 40 minutes
Total Marks: 60 (will be converted to 30).

CO Distribution:
CO1 - Answer any one out of two questions. There might be several sub-sections of each question such as 1a.(i),(ii) and so on. The questions will test your theoretical knowledge and concepts. It carries 10% of total marks.

CO2 - Answer any one out of two questions. There might be several sub-sections of each question such as 2a.(i),(ii) and so on. The questions will be a combination of derivations and theories both so it is essential to keep both theoretical concepts clear and practice derivations. It carries 30% of total marks.

CO3 - Answer any two out of three questions. There might be several sub-sections of each question such as 3a.(i),(ii), 3b(i),(ii) and so on. The questions will be purely based on maths and the structure is quite similar to what you have seen in quizzes, assignments, class examples. It carries 60% of total marks.

Note that skipping chapters is not an option since the questions are not made purely based on one chapter it is always a combination of multiple concepts.

Chapter-04

- **Fixed Point Iteration**

    - **Practice quiz and assignment questions for math**
    - **Contraction Mapping Theorem [no derivation]. Know the concept properly and why do we need it.**
    - **Convergence/ root found depends on two things - initial point consideration & function is converging or diverging**

- **Newton's Raphson Method**

- **Practice quiz and assignment and class examples for math [both with error bound and without error bound]**
- **Derivation - Newton's Method is super linear convergence. It is also an advantage for this method.**
- **Disadvantages related to turning point for Newton's Method?**
- **Secant Method - theory and maths.**

- **Bisection Method, Aitken Acceleration are not part of your final syllabus. Do not get confused by encountering them in concised notes/chapterwise notes.**

Chapter - 05

Triangular Method

- **No of. operations. -n^2**
- **Link: https://drive.google.com/drive/folders/1rLIPHxk1Y6hB7qc8PlWj-fXod5oSG453**
  Page 49 has triangular systems explained in details.
- **Source which I have discussed during class.**
- **5. Linear Equations**
- **Refer to pages 1,2 and 3 of Linear Equation 1.**
- **For Inverse Matrix, Gaussian Elimination, LU Decomposition going through class examples, quizzes and assignments should be sufficient.**

- **Why do we use LU Decomposition even though we have Gaussian Elimination?**
- **What is pivoting?**

Chapter - 06

- **What is an overdetermined system?**
- **How do we solve an overdetermined system? Two methods that I have discussed during class are Discrete Least Square and QR Decomposition.**
- **What is orthonormality? Practicing class examples and assignments should be sufficient.**
- **For Discrete Least Square, Gram Schmitt Process and QR Decomposition practice the maths shown in class, present in concise notes, assignment and quizz.**

Chapter - 07

- **Closed Newton's Cotes Formula with n = 1 / Trapezium Rule derivation, Closed Newton's Cotes Formula with n = 2 / Simpson's Rule derivation as shown in class and discussed in recording as well. It is essential to know the steps in details and also keep in mind the notations such as lagrange basis, weight function etc.**
- **Practice maths from class notes, practice sheet, assignment and quiz.**
- **Practice sheet** https://drive.google.com/file/d/15p4YcGKCeO7e-HdtmJX4xwKUDK6rcnu4/view?usp=sharing
- **Composite Newton's Cotes Formula [No derivation] always check notation for sub-intervals.**
- **% relative error related math as shown in class.**
  |approximate - actual | / |actual| * 100%